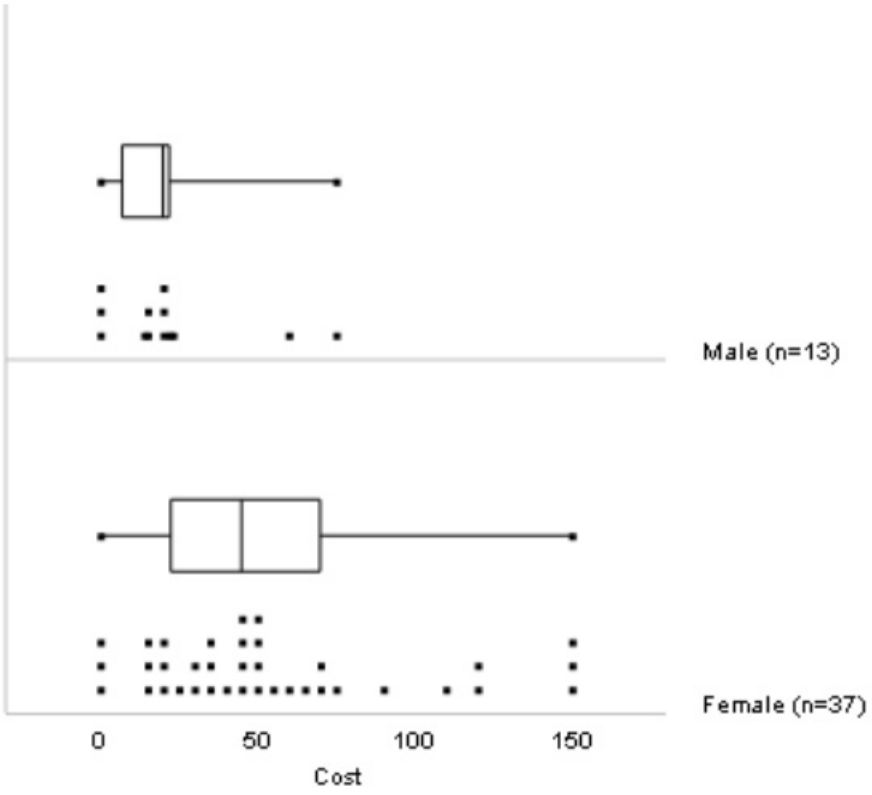
Solutions to Exploration 6.1A

1. Explanatory: Sex (categorical), Response: Haircut price (quantitative)
2. Observational study; no treatment was imposed by the investigator.
3. Neither
4. The center of the haircut distribution for women (≈ 50) is quite a bit larger than for men (≈ 20), Similarly the variability is substantially larger for women than it is for men. Both distributions show right skew, with a couple of extreme values for men occurring above $50, and a few for women ≈ $150.
5. a. Most people don't pay a lot for haircuts, but some people vary a whole lot, especially women who may be having more sophisticated haircuts (e.g., highlights).
   b. If people have a friend or family member cut their hair it will cost nothing, but as soon as you have a professional do it you are unlikely to pay less than ≈ $15.
6. Men: ≈ 20, Women: ≈ 50
7. Women; the data is much more spread out (larger variability).
8. a. Men: $n = 13$, mean = $21.85, SD = $22.14, Women: $n = 37$, mean = 54.05, SD = $41.61,
   b. Women (yes), c. Women (yes)
9. Yes, these data seem to show an association between sex and haircut price. Women tend to pay more for their haircuts than men.
10. a. Yes, women TEND to (statistical tendency) pay more than men – this is not always the case.
    b. 54.05 – 21.85 = $32.20
11. The means are still far apart, so while the small number of men is something to keep in mind, the difference in means is still large.
12. No, this is not a random sample of any population. While the sample may represent students at the university, or even all college students, we need to be pretty careful about generalizing since this is not a random sample, so there is no guarantee that the sample is representative.
13. Yes, probably, the difference in means is pretty big.
14. a. Men: 20, Women: 45, b. Women (yes), c. The medians are less than the means, which makes sense because the data is right skewed; skewness/extreme values will pull the mean in the direction of the extreme values; medians are not as affected by extreme values as the mean.
15. a. Women Q1 = 22.5
    b. Women Q3 = 70
    c. Men Q1 = 7, Q3 = 22.5

16.

	Minimum	Lower quartile	Median	Upper quartile	Maximum
Women's haircut prices	0	22.5	45	70	150
Men's haircut prices	0	7	20	22.5	75

17.



The women's haircut prices are more spread out compared to men's prices. This is supported by the 5-number summary in that the middle 50% of process (from Q1 to Q3) for women covers a much larger range than do the middle 50% of prices for men. Both distributions are skewed right as the max value is farther away from Q3 than the min value is from Q1.

18. IQR: Women:70 – 22.5 = 47.5; Men: 22.5 – 7 = 15.5
19. New SD = 16 (22.14 before), New IQR: 14 (15 before). The SD changed substantially more than the IQR with the removal of the "extreme" male haircut price of $75.
20. Analysis will depend on your class.

Solutions to Exploration 6.1B

1. The exact reading level for those < 3 and > 12 is not known, so 23 of the 63 patients don't have an exact reading level.
2. Patient median = 9; pamphlet median = 9
3. The medians are the same.
4. Not necessarily; the spread of reading levels needs to be considered, as does the shape of the distribution of reading levels for both the pamphlets and the patients.
5. 17/63 (27%) can't read even the simplest pamphlet.
6. Pamphlets need to err on the side of being readable to all patients.

Solutions to Exploration 6.2

1. What are the effects of not sleeping during the night on one's functioning? Could explore physical functioning and cognitive functioning. Could test these abilities the day after and then again several days after when the subjects are allowed full nights of sleep in those subsequent nights.
2. Explanatory: sleep-deprived/unrestricted (categorical); Response: improvement score (quantitative)
3. Experiment. In fact, it is a randomized experiment, because the researchers randomly assigned the subjects to the treatment groups.
4. Any of the following are acceptable:

   Null: There is no association between treatment group (unrestricted sleep/sleep-deprived) and improvement score.
   Alt: There is a positive association between treatment group (unrestricted sleep/sleep-deprived) and improvement score, such that the unrestricted sleep group tends to have better improvement scores.

   Null: the long-run average improvement score in the unrestricted sleep group is the same as the long-run average improvement score in the sleep-deprived group.
   Alt: The long-run average improvement score in the unrestricted sleep group is larger than the long-run average in the sleep-deprived group.

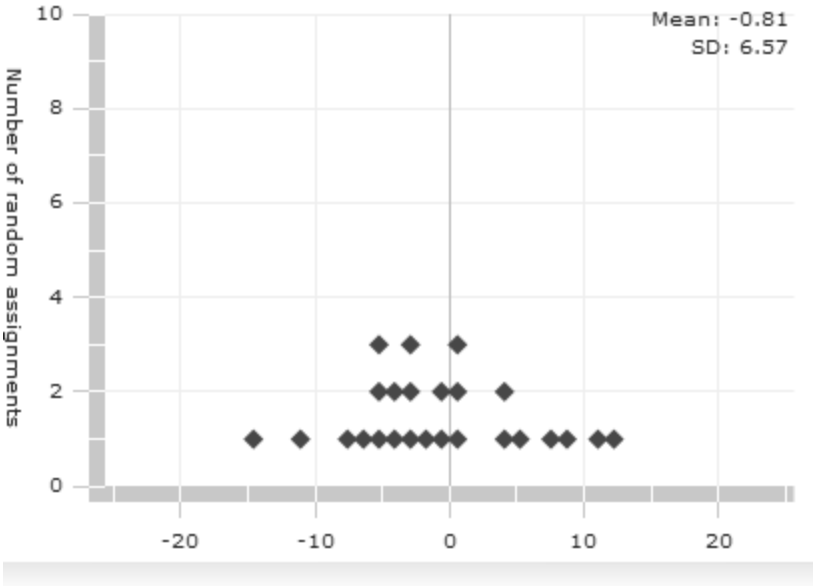
   Null: The difference in the long-run average improvement scores between the two groups is zero.
   Alt: The difference in the long-run average improvement scores (average unrestricted minus average sleep-deprived) is larger than zero.

   Null: $\mu_{\text{unrestricted}} = \mu_{\text{sleep-deprived}}$
   Alt: $\mu_{\text{unrestricted}} > \mu_{\text{sleep-deprived}}$
5.
   a. Unrestricted; those dots tend to be larger values than the sleep-deprived group.
   b. Unrestricted; there is more overall spread in the dots (range is larger).
   c. i. Unrestricted: sample size is 10, mean is 19.82, SD=14.73; ii. Deprived: sample size is 11, mean is 3.9, and SD=12.17.
   d. Unrestricted (19.82 > 3.9)
   e. Unrestricted (SD 14.73 > 12.17)
   f. 15.92

g. Yes, the difference in means seems large, especially relative to the overall variability (range) of the data.

6. It occurred just by chance, or there is really an effect of sleep deprivation.
7. In Chapter 5 we used playing cards where the color of the card represented the value of the response variable (black or red) and we shuffled and dealt the cards into two stacks (one for each explanatory variable group) to simulate the null hypothesis. By writing the observed values for the response variable (numbers) on the cards (instead of using colors), we can use a similar simulation strategy.
8. 15.92
9. a. 21

   b. The 21 different improvement scores. The remaining answers will vary, but you may end up with a dotplot like the one shown below (for a class of 25 students).



   c. Centered near 0; makes sense because null hypothesis says there is no difference in the group means.

   d. Nothing as large as the observed difference in mean (15.92) happened in the 25 simulations shown here; thus 15.92 is rarely occurring by chance.

10. a. The applet is shuffling the "dots" (improvement scores) and randomly putting them into two groups---this is simulating the null hypothesis of no effect of the group on the value of the improvement score b. Answers will vary, but we got – 4.24. This is not as large as 15.92. c. Yes, the answer changed. d. Yes, it changed again – almost every simulation will yield a different value for the difference in means. e. Generated 1000 simulations f. i. Each dot represents the difference in means from a single simulation. ii. Bell-shaped and centered at zero iii. The observed difference is in the tail. Didn't get 15.92 or larger very often g. Approximate p-value of 0.006 (answers will vary) h. The p-value 0.006 is the probability of observing 15.92 or larger assuming that sleep deprivation has no effect on improvement scores.
11. We have very strong evidence.
12. We have very strong evidence that sleep deprivation lowers the average improvement score.
13. a. SD = 7, 2SD = 14, so 15.92 ± 14 is (1.92, 29.92); b. I'm 95% confident that the long-run average improvement score is 1.92 to 29.92 higher with the unrestricted sleep treatment as opposed to the sleep-deprived treatment.
14. No. We won't be very comfortable generalizing this result to any population. The best we could do may be 18 – 25 year-old college students, but still no guarantee it's representative of that group.
15. Yes, they were randomly assigned, so the statistically significant finding suggests that the sleep deprivation may be causing the change in improvement scores.
16. a. Unrestricted: 16.55, Sleep-deprived: 4.5, Difference = 12.05, b. yes, bell-shaped and yes, centered at zero, c. Count samples above 12.05 gives p-value of approximately 0.018 (answers will vary) d. Yes, the p-value gives us strong evidence e. The means gave stronger evidence because the p-value was smaller.
17. It would be nice to have a sampling method that allowed us to generalize more broadly. It would also be a good idea to have at least 20 subjects in each group. Future studies could look into the potential effects of the way sleep-deprived individuals catch up on sleep, how long the effects of sleep-deprivation last and effects of sleep deprivation on other cognitive tasks. Expansion to other (more diverse) groups of individuals may also be warranted.

Solutions to Exploration 6.3

1. Explanatory: Sex (categorical); Response: Number of people (quantitative)
2. Random sampling only
3. Observational study – the researchers did not assign values of the explanatory variable.
4. Null: Among US adults, there is no relationship between a person's sex and the number of friends they have; Alt: Among US adults, there is a relationship between a person's sex and the number of friends they have.

   Null: The average number of friends for men in the US is the same as the average number of friends for women in the US. Alt: The average number of friends for men in the US is different than the average number of friends for women in the US.

5. The average number of close friends among all adult US men (μ_{men}). The average number of close friends among all adult US women (μ_{women}).
6. Null: $\mu_{men} = \mu_{women,}$ Alt: $\mu_{men} \neq \mu_{women}$
7. a. men: sample size = 654, mean = 1.86, SD = 1.78, women: sample size = 813, mean = 2.09, SD = 1.76. Women, $2.09 > 1.86$ (compare means)
   b. men $1.78 > 1.76$ (compare SDs)
   c. Right skewed – most people have a few close friends, but a few people have a large number.
   d. – 0.228
   e. It is the difference in the average number of friends (men minus women). In particular, $1.78 - 2.09 = -0.228$.
8. a. p-value = 0.014, b. The p-value of 0.014 is the probability of observing – 0.228 or more extreme (< -0.228 and > 0.228) assuming that there is no difference in average number of friends for males and females in the US population.
9. The p-value would get smaller.
10. Increasing the sample size will reduce the p-value because the variability of the null distribution will decrease.
11. Increasing the standard deviation will increase the variability of the null distribution and the p-value will increase.
12. The standardized statistic is $t = -2.45$ which means the observed difference in averages is 2.45 standard deviations below zero (the mean of the null distribution).
13. Small – it is < -2, meaning the observed statistic is more than 2 SDs away from the mean.

14. Symmetric, bell-shaped, and centered at zero.
15. Yes, the *t*-overlay matches the simulated *t*-distribution. This suggests the validity conditions will be met.
16. Very similar
17. Yes, the validity conditions are satisfied because the sample sizes are both well above twenty in each group (654 and 813 are both > 20) and the distributions are not strongly skewed.
18. a. Yes, the average number of close friends is different between US men and women.
    b. No, we are not saying that the number of friends one has is caused by one's sex, because this is not a randomized experiment.
    c. This result generalizes to all US adults because it is based on a random sample of all US adults.
19. a. The difference in the average number of friends between men and women in the US population.
    b. (– 0.4100,– 0.0454)
    c. No, it does not contain zero, implying that a difference of zero between the two means (men and women) is not plausible.
    d. Yes, we had evidence to reject the null hypothesis, which stated that the two group means were the same; the small p-value gave us evidence that it's not plausible that the two group means are the same in the population, based on the evidence we have in our sample.

    e. We are 95% confident that the mean number of close friends men have is smaller than the mean number of close friends that women have, by between 0.0454 and 0.4100.

    f. Increase because you are more sure you 'caught' the parameter in the interval.

    g. Decrease because with an increased sample size you have more precision.

    h. Increase because with increased variability you have decreased precision.

20. Number of friends was self-reported. Maybe find a more objective way to report number of close friends. Next steps could include different countries, to see if the same result is observed outside of the US. Could also compare elderly men and women, or on the opposite side of the spectrum young boys and girls.